

Component	#	Qty	Description	Purpose	Source / Notes	 Column 1
Raspberry Pi 5		1	High-performance single-board computer (2.4 GHz	High-level brain; handles computer vision sensor fu	Main Controller	£259.99
Huskylens 2 Camera		1	Edge-AI vision sensor	Localized color recognition and obstacle tracking.	Connected via Hardware I2C	£109.90
Ultrasonic Transducers (HC-SR04)		3	Range-finding distance sensors	Set up as a 180° triple array for obstacle detection ;	Mounted on custom LEGO brackets	£14.99
Metal-Geared DC Motors		2	High-torque micro DC motors	Provides rear-wheel traction and speed propulsion.	Drive Motors	£9.99
High-Torque Micro Servo		1	Metal-g geared steering actuator	Controls the front Ackermann steering mechanism.	Steering Actuator	£13.88
TB6612FNG		1	Dual H-Bridge motor driver	Safely interfaces the DC motors with the Raspberry	Motor Driver	£9.98
INIU 65W Power Bank		1	High-output USB-C power bank	Supplies stable continuous power to the Pi 5 and m	Main System Power	£46.99
XL6009 DC-DC Boost Converter		1	Adjustable step-up voltage regulator	Steps up voltage to supply the target voltage for the	Voltage Booster	£9.99
Resistors (1 kΩ & 2 kΩ)		2	Basic carbon film resistors	Forms voltage dividers to protect the Pi 5's 3.3V GF	Custom safety wiring	£9.99
LEGO Spike Essentials		1				£603.40
LEGO Structural Assembly		0	Beams frames pins and connectors	Serves as the main physical chassis and frame.	Sourced entirely from LEGO SPIKE boxes	
LEGO SPIKE Blue Wheels		4	56mm x 26mm high-grip rubber tires and blue hubs	Provides high-traction track contact and stable con	Sourced from LEGO SPIKE Prime	
LEGO Steering Linkages	—		Technic steering arms hubs and turnbuckles	Forms the physical Ackermann steering linkage.	Sourced from LEGO SPIKE boxes	
Total:						£1,089.10